



Racial and Ethnic Disparities in Outpatient Substance Use Disorder Treatment Episode Completion for Different Substances



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ABSTRACT

This study investigates how racial and ethnic disparities in treatment episode completion vary across different problem substances in an urban sample of 416,224 outpatient treatment discharges drawn from the 2011 U.S. Treatment Episode Dataset-Discharge (TEDS-D) data set. Fixed effects logistic regression is employed to test for the association of race and ethnicity with treatment episode completion for different substances of use while controlling for confounding demographic, socioeconomic, and geographic clustering factors. Results show that African Americans and Hispanics are less likely to complete a treatment episode than Whites, and that these disparities vary among users of different substances. For African Americans, this disparity is observed over all substances, but is particularly acute among users of alcohol and methamphetamine, substances for which African Americans generally have lower rates of use disorder as compared to Whites. For Hispanics, this disparity is driven primarily by users of heroin, for which Hispanics are only 75% as likely as Whites to complete a treatment episode. For users of cocaine and methamphetamine, there is no significant difference between Hispanics and Whites in the likelihood of treatment episode completion. These results contribute to emerging research on the mechanisms of substance use disorder treatment outcomes and highlight the need for culturally appropriate treatment programs to enhance treatment program retention and associated positive post-treatment outcomes.

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1. Introduction

1.1. The problem

Racial and ethnic minorities have experienced disparities in healthcare across a broad range of diseases and health problems, including substance use disorder (Healthy People, 2020, 2010; NIDA, 2008; Smedley, Stith, & Nelson, 2003). Minority populations are differentially affected by the consequences of substance use disorders in terms of incarceration, health problems, stigma, and violence (Amaro, Arevalo, Gonzalez, Szapocznik, & Iguchi, 2006; Boyd, Phillips, & Dorsey, 2003; Caetano, 2003; Iguchi, Bell, Ramchand, & Fain, 2005). While treatment for substance use disorders can ameliorate these consequences, particularly for those who complete appropriate treatment (Brorson, Arnevik, Rand-Hendriksen, & Duckert, 2013), research indicates that African Americans and Hispanics tend to have more barriers to accessing treatment services, lower utilization rates, and less satisfaction with treatment as compared to Whites (Alegria et al., 2006; Marsh, Cao, Guerrero, & Shin, 2009; Wells, Klap, Koike, & Sherbourne, 2001; Wu, Ringwalt, & Williams, 2003), though other studies have found few differences in the rates of service utilization among groups (Keyes et al., 2008; Schmidt, Greenfield, & Mulia, 2006). The purpose of this study

is to investigate racial and ethnic disparities in outpatient treatment episode completion, and then to examine how such disparities differ among different problem substances. We focus specifically on differences among African Americans, Hispanics, and Whites, as these are the three largest racial and ethnic groups in the U.S. and among those receiving treatment (SAMHSA, 2015).

1.2. Treatment completion literature

Though addiction is generally considered a chronic relapsing disease that may be most effectively managed with a continuum of care model (APA, 2013; Dennis & Scott, 2007), it is still important to recognize the importance of treatment completion for specific episodes of care. For those who do receive treatment services, retention is generally a critical prognostic indicator of positive post-treatment outcomes for all client groups (Brorson et al., 2013). Research has shown that those who complete a treatment episode tend to have better health, fewer relapses, fewer readmissions, less future criminal involvement, and higher levels of employment and wages, and to maintain longer term abstinence (Brewer, Catalano, Haggerty, Gaine, & Fleming, 1998; Brorson et al., 2013; Evans, Li, & Hser, 2009; Garnick, Lee, Horgan, & Acevedo, 2009; Messina, Wish, & Nemes, 2000; Stark, 1992; The TOPPS-II Interstate Cooperative Study Group, 2003; Zarkin, Dunlap, Bray, & Wechsberg, 2002). However, only about one in ten Americans 12 or older who need treatment for a substance use disorder receives it at a specialty facility.

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Outpatient treatment is by far the most common modality, and compared to residential treatment, which in general tends to have higher completion rates, minorities are overrepresented in outpatient settings (SAMHSA, 2013). Unfortunately, for those in outpatient treatment, the majority of clients tend not to complete it (McCaul, Svikis, & Moore, 2001; McHugh et al., 2013; Santonja-Gómez et al., 2010; Wickizer et al., 1994).

Research has identified a great many factors that increase or decrease the likelihood of completing a treatment episode for a substance use disorder (Brorson et al., 2013; Craig, 1985; Stark, 1992). Among the many client admission variables that have been identified in prior research as associated with completing a treatment episode are age, employment, education, sex, race, and ethnicity, as well as the client's specific primary problem substance, i.e. substance of choice. In general, older age, greater education, and being employed have been consistently associated with a greater likelihood of completing a treatment episode for substance use disorders (Brorson et al., 2013; Stark, 1992).

The literature is far less consistent concerning the influence of the client's substance of choice on treatment retention. For example, some studies have found that the use of cocaine increases the likelihood of early dropout from treatment (e.g., Fishman, Reynolds, & Riedel, 1999; McHugh et al., 2013; Veach, Remley, Kippers, & Sorg, 2000), yet other researchers have not found a significant association (e.g., King & Canada, 2004; Maglione, Chao, & Anglin, 2000). The same inconsistency of association with early dropout has been found in studies involving users of opiates and alcohol (e.g., Fishman et al., 1999; McKellar, Kelly, Harris, & Moos, 2006; Nellore & Ernst, 2004; Petry & Bickel, 2000; Veach et al., 2000). Generally when more than one drug of choice is incorporated into the research, variations in completion rates are evident (e.g., Luchansky, Krupski, & Stark, 2007; SAMHSA, 2013; Veach et al., 2000). However, the majority of the treatment completion research literature involves single site studies that tend to focus on a particular drug using population (Brorson et al., 2013), which prohibits direct comparisons of treatment completion rates for different substances of choice. For example, studies have focused on treatment retention and engagement among female outpatient alcoholics (Graff et al., 2009), incarcerated male methamphetamine users receiving prison-based treatment (Joe, Rowan-Szal, Greener, Simpson, & Vance, 2010), treatment resistant opiate dependent outpatients (McHugh et al., 2013), heroin dependent or marijuana dependent adolescents in outpatient treatment (Smyth, Fagan, & Kernan, 2012; White et al., 2004), and homeless veterans (Justus, Burling, & Weingardt, 2006), to cite just a few examples.

Inconsistencies are also prevalent in the research literature concerning racial and ethnic disparities in treatment episode completion. Many studies have found that African Americans and Hispanics are less likely to complete substance use disorder treatment compared to Whites (Arndt, Acion, & White, 2013; Bluthenthal, Jacobson, & Robinson, 2007; Cooper, MacMaster, & Rasch, 2010; Guerrero, Marsh, Khachikiana, Amaro, & Vega, 2013; Jacobson, Robinson, & Bluthenthal, 2007a), though others have not found similar disparities in treatment retention (Lowman & Le Fauve, 2003; Niv & Hser, 2006; Niv, Pham, & Hser, 2009; Stack, Cortina, Samples, Zapata, & Arcand, 2000; Veach et al., 2000). Perhaps one of the reasons for the inconsistency of these results is the importance of confounding and moderating characteristics that are accounted for in the research. For example, Jacobson, Robinson, and Bluthenthal (2007b) found that when controlling for economic disadvantage, differences in treatment completion rates between minority groups and Whites were greatly reduced, though they did not disappear. More recently, Arndt et al. (2013) found that disparities in treatment completion among Blacks, Hispanics, and Whites varied considerably by state.

Given that evidence suggests that treatment completion varies both by race/ethnicity and substance of choice, the prevalence of certain drug and alcohol problems within certain racial ethnic groups is also of interest. Data from the 2014 National Survey on Drug Use and Health (NSDUH) show some variations as well as similarities across types of substances used among adults 18 and over (SAMHSA, 2015). Past

month use of cocaine was 0.6% for Whites, and 0.7% for both Blacks and Hispanics, while for marijuana past month use was 10.8% for Blacks, 8.8% for Whites, and 6.5% for Hispanics. Past month binge and heavy alcohol use rates were 25.0% for Whites, 23.9% for Blacks, and 27.5% for Hispanics. Perhaps of greater importance than just prevalence of use is the extent of substance use disorders across groups. The prevalence of substance use disorders for illicit drugs was 2.3% for Whites and 2.6% for Hispanics, but 4.2% for Blacks. For alcohol use disorders, the prevalence for Whites, Blacks, and Hispanics was 6.8%, 6.5%, and 7.5%, respectively. Notably, racial and ethnic associations with particular substances may also be specific to particular regions or cities. For example, Bluthenthal et al. (2007) found that in Los Angeles County, Blacks were more likely to report using cocaine than other drugs, Whites were more likely to use amphetamines, and Hispanics were more likely to report using heroin. Guerrero, Marsh, Khachikiana, et al. (2013) found that in Texas heroin use was most common among Hispanics, whereas on the east coast of the U.S., cocaine use was found by some studies to be of higher prevalence among Latinos than any other group (Alegria et al., 2006; Amaro et al., 2006).

Few studies, however, have explicitly examined the intersection of race and ethnicity with substance of choice in studies of treatment completion. One of the few studies to examine this issue actually did not find that substance of choice predicted outpatient treatment retention when race was incorporated into their model (McCaul et al., 2001). However, this study had a limited sample size, categorized primary substance based on lifetime use criteria, and divided their sample into three very general categories—"alcohol only," "drug only," and "alcohol plus drugs." They did not differentiate among specific substances (e.g., cocaine; heroin) within the general category of "drugs" for their analyses. Klein, di Menza, Arfken, and Schuster (2002), on the other hand, did find interaction effects with minority status and the number of drug-related problems, though specific drug of choice was not included in their models.

This body of research supports the view that it is important to explore factors that moderate outcomes for different subgroups of clients (MacKinnon & Luecken, 2008). Indeed, the review of Brorson et al. (2013) echoes Wierzbicki and Pekarik's (1993) call for a shift in research away from simple demographic predictors of dropout (in the latter case applied to psychotherapy research) toward greater exploration of how such predictors may vary among population subgroups.

1.3. Present study

The present study investigates racial and ethnic group disparities in outpatient treatment episode completion (including ambulatory, intensive and non-intensive outpatient treatment) among adults over age 18 in the urban U.S. using data extracted from a national dataset, the Treatment Episode Dataset-Discharge (TEDS-D; SAMHSA, 2013). We focus our investigation on outpatient treatment given the predominance of this modality, especially for minority clients. In addition, we limit our analysis to treatment offered in urbanized areas because there are considerable differences in drug use patterns, perceived need for treatment, and the availability of treatment between urban and rural settings, especially for African Americans and Hispanics (Borders, Booth, Stewart, Cheney, & Curran, 2015; Metsch & McCoy, 1999; Perron, Gillespie, Alexander-Eitzman, & Delva, 2010; Vélez et al., 2008; Wang, Becker, & Fiellin, 2013).

We test three hypotheses. First, we hypothesize that African Americans and Hispanics in U.S. urbanized areas have a lower likelihood of treatment episode completion as compared to Whites. Second, we hypothesize that the probability of treatment episode completion is associated with substance of choice, where use of heroin, cocaine, marijuana, and methamphetamines is associated with a lower probability of treatment episode completion as compared to use of alcohol. Third, given the fact that the substance of choice tends to vary by racial and ethnic group, and that treatment episode completion varies by substance of choice, we hypothesize that the association between minority

group status and treatment episode completion will differ in strength across different problem substances.

2. Materials and methods

2.1. Data source

Data on substance use disorder treatment were extracted from the latest publicly available TEDS-D data set (SAMHSA, 2013), a compilation of data on discharges from substance use disorder treatment facilities across the U.S. which occurred in 2011. The data are collected by state-level substance use agencies, typically for facilities that receive public funding or for those which are licensed at the state level, and for which data are submitted to the federal government. While not all discharges from treatment facilities in the U.S. are included in the data, the data set captures a broad cross-section of discharges nationwide and can thus be considered a national sample (SAMHSA, 2013).

Because our focus is on urbanized areas, we only include TEDS-D discharge data from programs in metropolitan or micropolitan regions of the U.S., which are defined by the U.S. Census Bureau as Core Based Statistical Areas (CBSA). Each CBSA is composed of a county which contains an urbanized area of at least 10,000 persons (for a micropolitan area) or 50,000 people (for a metropolitan area), as well as adjacent counties which are seen to be economically dependent or interactive with the host county. Those not living in a Census-defined CBSA (i.e. discharges occurring in rural areas) were excluded from the study.

2.2. Variables

The following dichotomous variables were extracted from the TEDS-D data, where each variable was coded such that the presence of the characteristic was coded as one (1) and the absence coded as zero (0). The outcome variable, *Treatment Completion*, was defined as completing a treatment episode, in contrast to not completing a treatment episode due to leaving or being terminated from the program. The TEDS-D data set includes several values that indicate treatment episode non-completion, which include “Left against professional advice,” “Terminated by facility,” “Transferred to another treatment program or facility,” “Incarcerated,” “Death,” “Other,” and “Unknown.” Discharges with non-completion due to transfer to another program, incarceration, death, or for other reasons (“Other” or “Unknown”) were excluded from the analysis. Although we acknowledge that these excluded discharges may be viewed as a form of non-completion, we were most interested in contrasting what is clearly a known successful programmatic outcome (treatment episode completion) versus a negative programmatic outcome (Scalise, Berkel, & Van Whitlock, 2010), to the exclusion of ‘neutral’ outcomes such as transfer to another program. We note that the causal mechanism of treatment episode non-completion for those subjects who died or were incarcerated are qualitatively different from those who had the opportunity to complete a treatment episode but did not do so because they either left the program voluntarily or were terminated by the facility. We are interested here in understanding why a treatment episode may not have been completed among those who had the opportunity to do so, and therefore excluded those clients who did not have the opportunity to complete a treatment episode due to transfer, death, incarceration, or for other reasons. We also note that other researchers (Arndt et al., 2013; Sahker, Toussaint, Ramirez, Ali, & Arndt, 2015) who compared the use of different classifications of treatment non-completion (i.e., the consideration of transfers and incarceration as treatment non-completion outcomes) using the TEDS-D data found that it did not influence their results.

We included among our explanatory variables demographic variables indicating sex and age, which have been found to influence treatment completion rates in prior research. The variable *Male* encoded whether the sex of the subject was male as compared to female. The age of each subject discharged was recorded using the variables *Young*

(age 18–29), *Middle* (age 30–44), and *Old* (age >45). Race/ethnicity was coded using three mutually exclusive and exhaustive categories of *White* (not including Hispanics), *African American* (not including Hispanics), and *Hispanic*. Any individual identifying as Hispanic was classified as Hispanic no matter their racial identification. Clients with other race/ethnicity identities were excluded from the analysis as they represented a relatively low number of discharges.

As socioeconomic status has also been found to be associated with treatment episode completion, particularly for minority populations (Jacobson et al., 2007a, 2007b), we included two indicators of this variable contained in the TEDS-D dataset—client educational attainment and employment status. Regarding the former, the variable *High School* represents whether the subject completed high school or not. The variable *Unemployed* indicates whether the discharged subject is unemployed, as compared to being employed full or part time, or not in the workforce. Concerning our major variables of interest involving primary problem substance, we also encoded the substance of choice for each discharged subject, coded as *Alcohol*, *Cocaine* (including crack), *Marijuana* (including hashish), *Heroin* (including other opiates or synthetics), and *Methamphetamine* (or other amphetamines).

2.3. Analytic plan

We first compare treatment episode completion rates among different population subgroups, including subgroups defined by race/ethnicity and substance of choice, and employ the chi-square statistic to identify significant differences in treatment completion. We then implement several logistic regression models to test whether African American or Hispanic status is associated with treatment episode completion after controlling for demographic and socioeconomic factors, as well as for substance of choice. First, we regress race/ethnicity identity on treatment episode completion using the age, sex, socioeconomic status, and substance of choice variables as controls. This model takes the form

$$\ln(\hat{p}_i/1-\hat{p}_i) = \beta_0 + \beta_1 \text{Young}_i + \beta_2 \text{Middle}_i + \beta_3 \text{Male}_i \\ + \beta_4 \text{Black}_i + \beta_5 \text{Hispanic}_i + \beta_6 \text{HighSchool}_i \\ + \beta_7 \text{Unemployed}_i + \beta_8 \text{Cocaine}_i + \beta_9 \text{Marijuana}_i \\ + \beta_{10} \text{Heroin}_i + \beta_{11} \text{Methamphetamine}_i + \epsilon_i$$

where $\hat{p}_i$ is the estimated probability of treatment episode completion for discharge i , β is a parameter to be estimated, and ϵ is the error term. The variables indicating older age (45 years and over), White (non-Hispanic), and alcohol as the problem substance are left out as reference categories for the age, race/ethnicity, and substance of choice categorical variables, respectively.

Because there is evidence that treatment outcomes may cluster geographically (Arndt et al., 2013; Karriker-Jaffe, 2011), we decided to control for potential regional variation in treatment episode completion that was omitted from the initial model. We account for variations across urbanized areas using a fixed effects model, where dummy variables are employed to account for the identity of the CBSA within which each discharge is located. Finally, to investigate the role of race/ethnicity for different substances of choice, we fit separate fixed effects logistic regression equations for each problem substance, e.g. a model for users of cocaine, another model for marijuana users, and so on. Because we are making multiple comparisons across five models (i.e. one model for each problem substance), we report the significance values and confidence intervals associated with each odds ratio using the Bonferroni correction for ten pairwise comparisons. For all models, collinearity diagnostics, including variance inflation factor (VIF), were used to ensure that issues of collinearity were not problematic. Interpretation of the model results is complemented by reporting the frequencies and percentages of the primary substance of choice among Whites, Blacks, and Hispanics.

The effect of an independent variable on the likelihood of treatment episode completion may be assessed based on the odds ratio and its associated significance. However, because the sample size is relatively large ($N = 416,224$), even odds ratios near 1.00 may be highly significant. Such a result would suggest that while we may be confident that the observed effect of that variable on treatment completion reflects its effect among the population from which the sample is drawn, the magnitude of the effect itself may yet be trivial. While there is no strict rule as to what odds ratio value constitutes a ‘meaningful’ result, and qualitative descriptions may be used effectively to compare and characterize odds ratios, it may also be useful in this context to consider a conventional measure of effect size using the well-known Cohen’s d (Cohen, 1988), which distinguishes between small ($d = 0.2$), medium ($d = 0.5$), and large ($d = 0.8$) effect sizes. The odds ratio (OR) may be converted to Cohen’s d according to the formula $d = \sqrt{(\ln(OR))/1.81}$ (Chinn, 2000). Small, medium, and large Cohen’s d values correspond to odds ratios of 0.70, 0.41, and 0.24, respectively, when the variable is associated with a lower probability of treatment episode completion, and 1.43, 2.49, and 4.25, respectively, when associated with a higher probability. Thus, these odds ratio thresholds may be used as a guide for distinguishing between small, medium, and large effect sizes in the logistic regression results presented here, under circumstances when even trivial effect sizes (i.e. Cohen’s $d < 0.10$, or odds ratios between 0.83 and 1.19) are likely to be associated with high significance values.

3. Results

3.1. Sample characteristics

Descriptive statistics for each variable are provided in Table 1. Notably, approximately two thirds of the sample is male and slightly less than half the sample (47%) completed a treatment episode. This is slightly higher than the 44% outpatient treatment episode completion rate found in other studies utilizing TEDS-D data (Arndt et al., 2013; Sahker et al., 2015) and may indicate a higher rate of treatment episode completion in urban as compared to rural areas. Forty percent of the sample is unemployed, and alcohol is by far the most common substance problem (41% of all cases). Unadjusted treatment episode completion rates for different subgroups are also reported in Table 1, along with the chi-square statistic and associated significance value. Significant differences in completion rates were found for all groupings. Discharges of older age (45 years and over) have a higher completion rate than those of middle or younger age, though the difference in completion rate is only 2%. Males have a higher completion rate (48%) as

compared to females (43%). Treatment episode completion varies substantially by race/ethnicity, with Whites having the highest completion rate (50%) and African Americans the lowest (40%). Not surprisingly, treatment episode completion is associated with socioeconomic status, where high school attainment and employment are both associated with higher completion rates. Treatment episode completion also varies substantially among substance of choice, with alcohol having the highest completion rate (57%) and heroin the lowest (30%).

3.2. Predictors of treatment episode completion

Results of the logistic regression of treatment episode completion are reported in Table 2, where Model 1 reports the odds ratios for the explanatory variables, including the demographic, socioeconomic, and substance of choice characteristics of the clients. Collinearity diagnostics indicate the absence of collinearity among the explanatory variables, with all VIF values less than two. Results generally support the treatment completion patterns described above, where the probability of treatment episode completion is associated with older age, male sex, being White, higher socioeconomic status, and alcohol (as compared with other substances) as the substance of choice, with cocaine and heroin users having the least likelihood of completion.

A consideration of effect size, however, suggests that the effect of certain explanatory variables on treatment episode completion is of trivial magnitude. For example, though middle age subjects are significantly less likely to complete a treatment episode as compared to those aged 45 years and over, the odds ratio is 0.972 (Cohen’s $d = 0.02$), hardly a substantial difference in the probability of treatment completion. Contrast this with the effect of high school attainment, where a high school graduate is more than 1.3 (Cohen’s $d = 0.14$) times more likely to complete a treatment episode as compared to a non-high school graduate. The effect of unemployment is of a similar magnitude, where someone who is unemployed is only 79% (Cohen’s $d = 0.16$) as likely to complete a treatment episode as someone who is employed (or not in the labor force).

3.3. Racial and ethnic disparities in treatment episode completion

Our first hypothesis concerning the association of race and ethnicity with treatment episode completion is supported. However, while both African Americans and Hispanics are significantly less likely to complete a treatment episode as compared to Whites, the effect size is relatively trivial for Hispanics, who are 92% (Cohen’s $d = 0.04$) as likely to complete a treatment episode as are Whites. African Americans, on the

Table 1
Frequencies and treatment completion rates for outcome and explanatory variables ($N = 416,224$).

Variable	Values	N	% N	% Compl. Treatment	Pearson χ^2
Treatment Completion	Completed	193,670	47%	–	–
	Not Completed	222,554	53%	–	–
Age	Young (18–29)	164,011	39%	46%	118.882***
	Middle (30–44)	149,964	36%	46%	
	Old (>45)	102,249	25%	48%	
Sex	Male	277,236	67%	48%	960.071***
	Female	138,988	33%	43%	
Race	White (not Hispanic)	245,209	59%	50%	2,813.112***
	African American (not Hispanic)	106,235	25%	40%	
	Hispanic	64,780	16%	47%	
Educational Attainment	Completed High School	287,577	69%	49%	2,617.232***
	Not Completed High School	128,647	31%	41%	
Employment	Unemployed	164,755	40%	42%	2,065.389***
	Not Unemployed	251,469	60%	49%	
Primary Problem Substance	Alcohol	170,598	41%	57%	19020.273***
	Cocaine/Crack	39,834	10%	39%	
	Marijuana/Hashish	89,419	21%	46%	
	Heroin	89,294	21%	30%	
	Methamphetamine	27,079	7%	52%	

*** $p < .005$.

Table 2Results of logistic regression models of treatment completion ($N = 416,224$).

	Variable	Model 1 (No Fixed Effects)	Model 2 (Fixed Effects)
Age	Young (18–29)	0.972*** (0.955–0.989)	0.922*** (0.906–0.939)
	Middle (30–44)	0.972*** (0.956–0.988)	0.947*** (0.931–0.963)
Sex	Male	1.157*** (1.142–1.173)	1.195*** (1.178–1.212)
Race/Ethnicity	African American	0.687*** (0.676–0.698)	0.771*** (0.758–0.785)
	Hispanic	0.924*** (0.907–0.941)	0.971*** (0.952–0.991)
Socioeconomic Status	Completed High School	1.310*** (1.291–1.328)	1.318*** (1.299–1.337)
	Unemployed	0.793*** (0.782–0.803)	0.755*** (0.745–0.766)
Primary Problem Substance	Cocaine	0.554*** (0.541–0.566)	0.560*** (0.547–0.575)
	Marijuana	0.742*** (0.729–0.756)	0.753*** (0.739–0.767)
	Heroin	0.325*** (0.319–0.330)	0.322*** (0.316–0.328)
	Methamphetamine	0.843*** (0.822–0.866)	0.730*** (0.709–0.752)

Notes: Odds ratios are reported; 95% confidence intervals are in parentheses.

*** $p < 0.005$.

other hand, are only 69% (Cohen's $d = 0.21$) as likely to complete a treatment episode as Whites. Put another way, Whites are nearly one and a half times more likely to complete a treatment episode as compared to African Americans, after controlling for the influence of age, sex, socioeconomic status, and substance of choice.

3.4. Substance of choice and treatment episode completion

Our second hypothesis regarding the association of substance of choice with treatment episode completion is also supported. As compared to alcohol users, users of cocaine, marijuana, or heroin were less likely to complete a treatment episode, with effect sizes that range between small and medium. The difference in treatment completion is particularly strong between alcohol and cocaine users and between alcohol and heroin users. Alcohol users are nearly twice as likely to complete a treatment episode compared to cocaine users (Cohen's $d = 0.33$), and are more than three times as likely to complete a treatment episode compared to heroin users (Cohen's $d = 0.62$). Methamphetamine users were also less likely to complete a treatment episode compared to alcohol users, though the effect size was trivial (Cohen's $d = 0.09$).

3.5. Predictors of treatment episode completion, controlling for geographic clustering

The fixed effects model is reported in Table 2, Model 2. Odds ratios and significance are not reported for the CBSA dummy variables for the sake of brevity (there are 244 CBSAs represented in the data set). Odds ratios for the explanatory variables remain largely the same in terms of direction, magnitude, and significance from Model 1 to Model 2. The effect sizes for both the African American and Hispanic variables decline slightly, whereas the effect size for the methamphetamine variable increases slightly. This suggests that whatever the effects of CBSA characteristics on treatment episode completion may be, they do not explain the relationships among the discharge-level explanatory variables and treatment completion as shown in Model 1.

3.6. Racial and ethnic disparities in treatment episode completion for different substances

Table 3 reports the results of the regression models for each problem substance: alcohol (Model 3), cocaine (Model 4), marijuana (Model 5), heroin (Model 6), and methamphetamine (Model 7). One can observe whether the confidence intervals of an explanatory variable for different models overlap – if they do not overlap, this indicates a significant difference between the two problem substances in the association of that explanatory variable with treatment episode completion. For instance, while Model 2 indicates that being less than 30 years of age generally decreases the likelihood of treatment episode completion as compared to those at least 45 years of age over the entire sample, the effect is particularly acute for users of methamphetamine and marijuana, significantly less so for users of heroin, and not significant for users of alcohol. Conversely, among alcohol users, men are more than 1.3 times more likely to complete a treatment episode as compared to women. However, sex is not significantly associated with treatment episode completion for users of methamphetamine or heroin. On the other hand, the effect of completing high school on treatment episode completion is fairly consistent in direction across substance of choice, though the effect is significantly stronger for marijuana and alcohol users as compared to users of cocaine, heroin, and methamphetamines.

These results support our third hypothesis regarding the variation in the association of race and ethnicity with treatment episode completion across different problem substances. For instance, while African Americans are significantly less likely than Whites to complete a treatment episode across all types of substances, the disparity is significantly stronger for users of alcohol as compared to users of cocaine, marijuana, and heroin. The disparity in treatment episode completion between Whites and Blacks is also quite strong for users of methamphetamines, significantly more so than for users of heroin. Interestingly, alcohol and methamphetamines are the substances which African Americans are less likely to use as compared to Whites and Hispanics (in this dataset). Table 4 shows the number and proportion of Whites, African Americans, and Hispanics in the dataset by primary problem substance. Notice that

Table 3

Results of fixed effects logistic regression models of treatment completion for different substances of choice.

Variable	Model 3 (Alcohol) $N = 170,598$	Model 4 (Cocaine) $N = 39,834$	Model 5 (Marij.) $N = 89,419$	Model 6 (Heroin) $N = 89,294$	Model 7 (Meth.) $N = 27,079$
Young (18–29)	1.024 (0.987–1.063)	0.788*** (0.718–0.866)	0.741*** (0.685–0.802)	0.922*** (0.866–0.982)	0.719*** (0.644–0.804)
Middle (30–44)	0.966 (0.933–1.001)	0.849*** (0.792–0.909)	0.885*** (0.814–0.961)	0.931*** (0.876–0.989)	0.820*** (0.739–0.910)
Male	1.330*** (1.288–1.373)	1.186*** (1.113–1.264)	1.193*** (1.141–1.248)	1.012 (0.968–1.058)	1.008 (0.936–1.086)
African American	0.707*** (0.680–0.735)	0.823*** (0.766–0.884)	0.777*** (0.741–0.815)	0.865*** (0.805–0.929)	0.662*** (0.548–0.801)
Hispanic	1.112*** (1.061–1.165)	0.974 (0.878–1.079)	0.904*** (0.849–0.963)	0.750*** (0.698–0.806)	0.996 (0.904–1.097)
Completed High School	1.373*** (1.326–1.421)	1.124*** (1.052–1.200)	1.413*** (1.356–1.472)	1.188*** (1.131–1.248)	1.223*** (1.132–1.321)
Unemployed	0.688*** (0.667–0.710)	0.886*** (0.830–0.945)	0.763*** (0.733–0.795)	0.875*** (0.835–0.916)	0.841*** (0.781–0.906)

Notes: Odds ratios are reported; Bonferroni corrected 95% confidence intervals are in parentheses.

** $p < 0.01$.*** $p < 0.005$ (Bonferroni corrected).

Table 4
Frequencies and percentages of primary problem substance by race/ethnicity (N = 416,224).

Substance	White	African American	Hispanic
Alcohol	112,367 (46%)	34,069 (32%)	24,162 (37%)
Cocaine	15,322 (6%)	18,957 (18%)	5555 (9%)
Marijuana	35,865 (15%)	38,166 (36%)	15,388 (24%)
Heroin	62,267 (25%)	13,944 (13%)	13,083 (20%)
Methamphetamine	245,209 (8%)	1099 (1%)	6592 (10%)
Total	245,209 (100%)	106,235 (100%)	64,780 (100%)

while alcohol is the most common problem substance among Whites (46%), only 32% of African Americans report it as their primary problem substance. Methamphetamine is the drug of choice almost exclusively for Hispanics and Whites. Interestingly, methamphetamine is the substance of choice for only 1% of African Americans who entered outpatient treatment in this sample.

One can also observe differences in the effect of being Hispanic on treatment episode completion for different substances. Being Hispanic suppresses the likelihood of treatment episode completion compared to Whites. However, this effect is, in fact, the opposite for those whose primary problem substance is alcohol, for which Hispanics have a significantly *higher* likelihood of treatment episode completion (though the effect size can be considered trivial, Cohen's *d* = 0.06). For marijuana, being Hispanic as compared to being White is associated with lower treatment episode completion (again, with a trivial effect size, Cohen's *d* = 0.06), whereas for cocaine and methamphetamine, being Hispanic (compared to Whites) has no significant effect on the likelihood of treatment episode completion.

4. Discussion

To our knowledge, this study is the first national level analysis to provide evidence that racial and ethnic disparities in urban outpatient substance use disorder treatment episode completion vary across substance of choice. While Hispanics and African Americans have generally lower treatment episode completion rates as compared to Whites, these disparities vary, and in some cases actually disappear, depending on the particular problem substance. Overall, across all substances, African Americans are only 69% (Cohen's *d* = 0.21) as likely to complete a treatment episode as Whites, while Hispanics are 92% (Cohen's *d* = 0.04) less likely to do so, after controlling for age, sex, socioeconomic status, and substance of choice. When controlling for geographic clustering, these patterns are generally similar, but the disparities in treatment episode completion are attenuated, with African Americans and Hispanics being 77% (Cohen's *d* = 0.14) and 97% (Cohen's *d* = 0.02) as likely to complete a treatment episode, respectively, as Whites.

Our results concerning treatment episode completion disparities between African Americans and Whites are generally consistent with most of the prior research that has examined this issue (Arndt et al., 2013; Cooper et al., 2010; Jacobson et al., 2007a). Our results regarding Hispanics and Whites are similar to several studies (Arroyo, Westerberg, & Tonigan, 1998; Niv et al., 2009; Tonigan, 2003), but contrast with other research that has identified greater differences in treatment episode completion (Bluthenthal et al., 2007; Guerrero, Marsh, Khachikiana, et al., 2013; Jacobson et al., 2007b; Milligan, Nich, & Carroll, 2004). Some of this discrepancy may be due to our focus on substance of choice since two of the prior studies that did not find disparities focused on alcohol treatment (Arroyo et al., 1998; Tonigan, Miller, Juarez, & Villanueva, 2002). Our finding that racial and ethnic group disparities in treatment episode completion vary by substance of choice is consistent with one of the few prior studies that specifically examined treatment completion and the interaction of race/ethnicity and substance of choice (Guerrero, Marsh, Duan, et al., 2013).

Given the disparity in treatment episode completion for minorities, particularly African Americans, as compared to Whites, as well as the

affinity of certain racial/ethnic groups for certain problem substances, it is fair to inquire whether observed racial and ethnic disparities in treatment episode completion are being driven by the lower completion rates associated with certain problem substances. For Hispanics, this indeed appears to be the case, at least to some extent, as the disparity observed among the full sample is driven largely by the fact that Hispanics are 75% (Cohen's *d* = 0.16) as likely to complete a treatment episode for heroin use as compared to Whites. Of note also is that among users of marijuana, Hispanics are also less likely to complete a treatment episode. While the effect size is perhaps trivial (Cohen's *d* = 0.06), a greater proportion of Hispanic clients enter outpatient treatment for marijuana use as compared to Whites (24% versus 15%); thus, the effect impacts a greater share of the Hispanic population.

The disparity in treatment episode completion between African Americans and Whites, however, does not appear to be driven by a particular substance of choice, even though African Americans are more likely to receive outpatient treatment (in our sample) for problems with cocaine and marijuana, and Whites are more likely to be in treatment for problems with alcohol, heroin, and methamphetamine. Rather, African Americans are less likely to complete a treatment episode compared to Whites across all substances of choice, although the disparity varies in magnitude among different problem substances. For heroin and cocaine users, the level of disparity in treatment episode completion between African Americans and Whites is somewhat weaker than for methamphetamine or alcohol, ranging between 87% (Cohen's *d* = 0.08) as likely to complete a treatment episode for African American heroin users down to 66% (Cohen's *d* = 0.23) for methamphetamine users. Only with heroin was the disparity in treatment episode completion with Whites greater for Hispanic clients than for African Americans.

One potential interpretation for the disparity in treatment episode completion between African American and White users of methamphetamines, and perhaps alcohol, is the negative effect of social isolation that may occur for African Americans in a treatment setting where the majority of clients are from a different social, economic, and/or cultural background. As Table 4 indicates, African Americans are less likely than Whites to be in treatment for alcohol use and substantially less so for amphetamine use, and thus African Americans may be particularly prone to social isolation among other methamphetamine and alcohol users in these treatment settings. Evidence indicates that travel to a location with a different socio-cultural context than one's home for outpatient substance use disorder treatment is associated with several negative outcomes, including relapse, anxiety, and treatment dropout (Mennis, Stahler, & Baron, 2012; Oser & Harp, 2015). Having to travel longer distances to treatment may also suppress the likelihood of treatment episode completion, and such accessibility may differ substantially by race and ethnicity (Guerrero & Kao, 2013; Guerrero, Kao, & Perron, 2013; Mennis et al., 2012). Such evidence reinforces the importance of developing and providing culturally relevant and tailored treatment interventions for minority populations that take into account differences in cultural and linguistic backgrounds (eg., Alegría et al., 2006; Amaro et al., 2006; Guerrero, Marsh, Khachikiana, et al., 2013; Lowman & Le Fauve, 2003; Smedley et al., 2003). However, culturally relevant interventions and research should also consider substance of choice as a potentially important component of socio-cultural context. It may be that minorities in treatment for substances which are less commonly used within their socio-cultural group are more vulnerable to the negative effects of social discordance within the treatment setting and, consequently, treatment episode non-completion.

The reasons for disparities in treatment episode completion are multi-determined and complex, and our focus on controlling for a limited number of demographic variables and geographic location at the city level in order to examine the interaction of treatment episode completion disparities and substance of choice leaves out many other possible factors that influence these differences in outcome. Prior research has identified a wide array of service and individual factors that also play a key role in racial and ethnic disparities in treatment outcomes

for substance use disorders. For example, a variety of studies have indicated that minority groups may have less access to treatment, experience greater delays in entering treatment, have a higher degree of unmet ancillary service needs (e.g., vocational training, health services, housing, etc.), report less satisfaction with treatment (including discrimination among service providers), have greater cultural barriers to service utilization (e.g., language proficiency, and lack of culturally sensitive treatment), experience greater health and mental health problems, have more criminal justice involvement, are exposed to higher levels of psychosocial stressors, are at increased risk for homelessness and residential instability, and have a greater severity of substance use disorder, all of which may influence an individual's treatment experience and the likelihood of treatment retention and completion (Alegria et al., 2006; Appel, Ellison, Jansky, & Oldak, 2004; Guerrero, Campos, Urada, & Yang, 2012; Guerrero, Marsh, Khachikiana, et al., 2013; Jacobson et al., 2007a; Marsh et al., 2009; Niv et al., 2009; Tonigan, 2003; Vega & Sribney, 2005; Wells et al., 2001; Wu et al., 2003).

Most of these studies rely on admission characteristics of clients (as we did) to examine outcomes. However, more research needs to take into account the interaction of client and program factors, as well as local contextual variables (Mennis et al., 2012; Stahler, Mennis, Cotlar, & Baron, 2009; Stahler et al., 2007) to improve our understanding of why clients, particularly ethnic and racial minorities, drop out or complete a treatment episode. In addition, combining quantitative and qualitative approaches for investigating treatment completion and dropout may yield more programmatically relevant information than any single approach (Stahler & Cohen, 2000; Stahler, Cohen, Greene, Shipley, & Bartelt, 1995; Stahler, Cohen, Shipley, & Bartelt, 1993; Stahler et al., 1993).

Finally, we also found that other client characteristics besides racial and ethnic identity varied in their relationship with treatment episode completion according to substance of choice. Consider, for example, older age, which has been a very consistent predictor of treatment completion in prior research (e.g., Brorson et al., 2013; McKellar et al., 2006). Our analytical results over the entire sample indicate, however, that the effect of age on treatment episode completion, while observable, is trivial (the equivalent Cohen's *d* values for the younger and middle age odds ratios reported for Model 2 in Table 2 are 0.04 and 0.03, respectively). Perhaps more importantly, however, the effect of age differed substantially among different problem substances. For those seeking treatment for alcohol use disorder, there was no difference in treatment episode completion between the youngest adult clients compared to the oldest group, but younger individuals were more than 21% to 29% (Cohen's *d* values ranging between 0.12 and 0.19) less likely to complete treatment if cocaine, marijuana, or methamphetamine was the substance of choice. Similarly, males were overall more likely to complete a treatment episode as compared to females, but there were no differences between men and women among heroin and methamphetamine users.

This study has a number of limitations. The data were extracted from a large public survey based on state administrative databases and therefore there may be inconsistent reporting in survey responses. This may be especially true for our primary outcome variable, treatment episode completion, as the criteria and definitions of "successful completion" may vary across programs. We also acknowledge that analyses of other treatment non-completion outcomes besides dropout or termination, such as incarceration or death, may yield different results, particularly given the potential association between incarceration and race and ethnicity. In addition, treatment episode completion is only one indicator of a successful outcome; other outcomes such as abstinence, employment, relapse, as well as other areas of client functioning, were not included. We used a limited number of admission variables to focus on our primary variables of interest, but there are clearly many other important variables of interest that have been shown to influence treatment outcomes that we did not include, such as referral source, criminal justice involvement, recent drug use, living situation (i.e. homelessness), history of mental problems, and prior treatment episodes. In addition, we limited our sample to only outpatient programs

in urbanized areas. It is unknown how our findings would translate to discharges from programs in rural areas of the U.S.

We should also note that minority groups as characterized in this large data set actually represent heterogeneous groups within this global classification. As Guerrero, Cepeda, Duan, and Kim (2012) and Guerrero, Marsh, Khachikiana, et al. (2013) have discussed, substance use patterns, barriers to accessibility, service utilization, and treatment completion all vary considerably among Hispanic subgroups. Finally, we employ fixed effects logistic regression in this study. While this approach is intended to control for metropolitan-based clustering in treatment episode completion so as to focus on issues of race/ethnicity and substance of choice, such clustering is, itself, also of interest. Methods analogous to the state-based investigations of Arndt et al. (2013) would be beneficial to investigate the form of such metropolitan-level variation, and multilevel modeling may be used to test hypotheses regarding regional socioeconomic or policy-based mechanisms of geographic variation in treatment episode completion.

5. Conclusion

As the Affordable Health Care Act continues to be implemented, with the resulting expansion of health insurance coverage for disadvantaged populations, minorities may have greater access to substance use disorder treatment. Given the vast literature showing the importance of treatment retention and treatment episode completion in terms of its effect on such longer term post-treatment outcomes of abstinence, employment, reduced criminal justice involvement, and health, improving our understanding of minority disparities in treatment completion becomes particularly important. Our research contributes to this understanding by highlighting how racial and ethnic disparities in outpatient treatment episode completion may differ across substance of choice. While disparities in treatment episode completion may place minorities at a disadvantage for long term sobriety and recovery, this issue needs to be examined through a more refined lens that includes the specific problem substance. Increasing treatment retention rates among minorities demands understanding not only why racial and ethnic disparities in treatment episode completion occur generally, but also the forces driving the variation in such disparities across different problem substances.

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